Publication List

Published/Accepted

1 - A (Green) Switch in Time Saves Nine: Assessing the Environmental Damage of the European Truck Cartel

- **Author List:** Ilona Dielen (ERUDITE, Univ. Paris-Est Créteil), Patrice Bougette (CNRS, Univ. Côte d'Azur), Christophe Charlier (CNRS, Univ. Côte d'Azur)
- **Journal:** Environemental and Ressource Economics (ISSN: 0924-6460), Vol. 89, article number 19 (2026)
- **DOI:** 10.1007/s10640-026-01073-6
- **Abstract:** This study examines how the cartel of European truck manufacturers coordinated the timing of compliance with emission standards, generating additional air pollution without violating environmental regulations. Although firms formally complied with environmental law, collusion restricted competition over cleaner technologies, highlighting that anticompetitive agreements can have significant environmental and health consequences. First, we quantify the volume of particulate emissions attributable to cartel behavior by constructing two plausible counterfactual scenarios for truck fleet composition, identifying substantial excess emissions of approximately 119 thousand tonnes of fine particulate matter ($PM_{2.5}$). Second, we estimate the health impact of traffic-related $PM_{2.5}$ emissions on infant respiratory outcomes using a panel of 199 European subregions observed over an 18-year period. To address endogeneity concerns, we exploit exogenous variation in EURO emission standards through a shift-share instrumental-variable strategy. The resulting elasticity allows us to compute the number of infant respiratory hospital admissions attributable to the cartel under counterfactual competitive conditions. We estimate that earlier, competition-driven adoption of cleaner technologies could have reduced average yearly infant hospital admissions by 12–18 cases per 1000 births at the NUTS 2 level.

2 - Impact Assessment of the "Damages Directive": Abuse of Dominance Cases

- **Author List:** Ilona Dielen (ERUDITE, Univ. Paris-Est Créteil), Jeanne Mouton (European Commission, DMA Unit)
- **Journal:** European Journal of Law and Economics (ISSN 0929-1261), [forthcoming] and referenced as GREDEG Working Paper, 2025-42
- **Abstract:** This study assesses the effect of Directive 2014/104/EU (the "Damages Directive") on the outcome of private actions for damages following an abuse of a dominant position. Using an original dataset of 194 cases decided in France, Italy, and the United Kingdom between 1988 and 2022, we implement a difference-in-differences design in which stand-alone damages actions, those most exposed to the Directive's procedural and evidentiary reforms, form the treated group. Conditional on a case being litigated, the adoption of the Directive is associated with an increase of 30.8 percentage points in the probability of a favourable outcome for treated cases. Furthermore, the estimate should be read as an average institutional effect on the likelihood of a favourable judicial outcome across three heterogeneous legal systems. The effect is robust to alternative definitions of the treated group and to functional form, and placebo tests reveal no differential pre-reform trend. In the cross-country analysis, France and the United Kingdom carry more than Italy. The estimate remains positive, but less precise, when treatment is dated from national transposition rather than adoption.

Under Review

3 - Is Green the New Black? Remote Work and the Rising Value of Natural Amenities

- **Author List:** Ilona Dielen (ERUDITE, Univ. Paris-Est Créteil), Catherine Laffineur (CNRS, Univ. Côte d'Azur)
- **Journal:** Regional Studies (ISSN: 0034-3404), under review since june 2025
- **Abstract:** Remote work has altered the relationship between the workplace and the place of residence, with consequences for the geography of housing demand that are as yet poorly understood. Using a municipality-by-year panel for metropolitan France (2018–2023), we ask whether remote-worker migration has raised housing prices and whether natural amenities have shaped where. We identify remote workers from matched employer–employee records as employees in occupations that can be done remotely who live more than 100 km from their workplace, capturing realized rather than potential remote work. Our shift-share instrument combines pre-pandemic bilateral migration links with the national, occupation-specific rise of long-distance remote work, and we conduct exposure-robust inference following Adão, Kolesár and Morales (2019). Instrumented inflows raise price growth only after 2020, accounting for about 18% of national post-2020 price growth. The effect is concentrated in amenity-rich areas rather than spread uniformly across the territory. It is large and significant in rural municipalities but not in urban ones, roughly twice as large on the coast, and largest where rural and coastal endowments coincide. Remote work thus emerges as a new spatial sorting mechanism shaped by environmental amenities.

4 - **Waiting Together: Cartels and the Timing of Environmental Standards**

- **Author List:** Patrice Bougette (CNRS, Univ. Côte d'Azur), Christophe Charlier (CNRS, Univ. Côte d'Azur), Ilona Dielen (ERUDITE, Univ. Paris-Est Créteil)
- **Status:** Revue d'Economie Industrielle "Environment & Competition" Special Issue, under review since october 2026
- **Abstract:** This paper analyzes a duopoly's incentives to adopt a mandatory environmental standard ahead of a known compliance date, and the effect of a cartel on this timing. Drawing on stylized facts from the European truck cartel, we combine the timing game of Reinganum (1981) with a vertical differentiation model à la Crampes and Hollander (1985) and show that: (i) under competition, early adoption yields the pioneer a differentiation rent, and adoption is staggered; (ii) the preemption race dissipates this rent, so that, when brand loyalty is strong, both firms prefer to agree to adopt only at the legal deadline; (iii) this cartel raises emissions by more the larger the emissions gap between standards, and worsens an adoption delay that already exists under competition. The model provides an equilibrium counterfactual for assessing the environmental damage of cartels.

Advanced Working paper

5 - **Air Pollution Cost Analysis: Predicting Social Health Expenditures (APOCALYPSE)**

- **Author List:** I. Bagayev (ERUDITE, Univ. Paris-Est Créteil), F.-O. Baudot (CNAM, ERUDITE, Univ. Paris-Est Créteil), I. Dielen (ERUDITE, Univ. Paris-Est Créteil), N. Gouaud-Gosteau (CNAM, ERUDITE, Univ. Paris-Est Créteil), and J. Lochard (ERUDITE, Univ. Paris-Est Créteil)
- **Status:** DREES report forthcoming, working paper ready to submission in fall 26
- **Abstract:** This paper provides the first nationwide, individual-level causal estimates of the effect of air pollution on health spending in France, combining 132 million beneficiary-months from the ESND (*Echantillon du Système National des Données de Santé*), a representative 2% sample of the French health insurance database, with weather-instrumented concentrations of $PM_{2.5}$, NO_2 and O_3 over 2014–2021. We find that a 10% increase in monthly $PM_{2.5}$ concentration (approximately 1 $\mu g/m^3$) raises total reimbursed healthcare expenditure by 0.70%; the corresponding figure for NO_2 is 0.55%, whereas the elasticity for O_3 ranges between 0.34 and 0.46%. Accounting for ozone jointly with primary pollutants is essential: omitting it substantially understates the effect of $PM_{2.5}$ and NO_2, particularly in urban areas, reinforcing the case against single-pollutant designs. The expenditure response is largest among children aged 5 to 14, who are around three times more sensitive than adults of working age. How people respond to a pollution shock also depends on their financial access to healthcare: those covered by means-tested schemes (CMU-C, ACS, C2S) may be both more vulnerable to pollution and, with no out-of-pocket costs, better able to seek reimbursed care. Yet the poorest municipalities, where the out-of-pocket barrier remains, respond least despite being the most exposed, suggesting that part of the health burden of pollution does not translate into reimbursed care.

6 - **Remote Work and Local Air Pollution: Evidence of a Post-2020 Rebound Effect**

- **Author List:** Ilona Dielen (ERUDITE, Univ. Paris-Est Créteil)
- **Status:** Manuscript under editing, working paper ready to submission in winter 26/27
- **Abstract:**The rapid expansion of remote work following the COVID-19 pandemic has reshaped residential mobility, with potentially important environmental consequences. This paper investigates whether the post-pandemic relocation of remote workers affected local air pollution in France. Using administrative data on remote workers and annual concentrations of PM2.5, PM10, and O3 for 8,548 municipalities over 2019–2023, I examine the relationship between remote worker inflows and local pollution. To address the endogenous spatial distribution of remote workers, I use a shift-share instrumental-variable strategy based on pre-pandemic occupational exposure to remote work and the subsequent expansion of remote employment. The results show that the redistribution of remote workers is associated with higher particulate matter concentrations and lower ozone concentrations, with heterogeneous effects across municipalities. These findings suggest that the environmental consequences of remote work depend not only on reduced commuting but also on the spatial redistribution of workers and the local rebound effects associated with residential relocation.